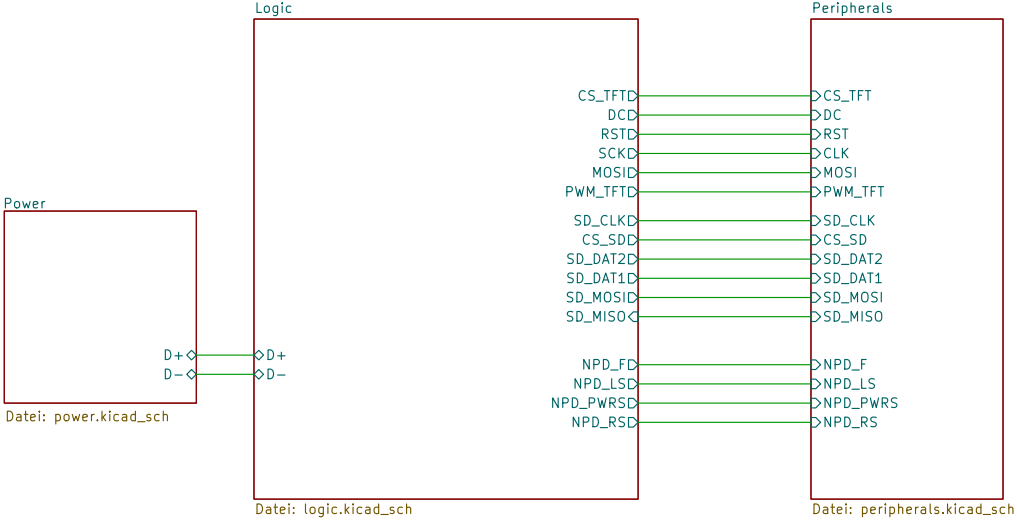
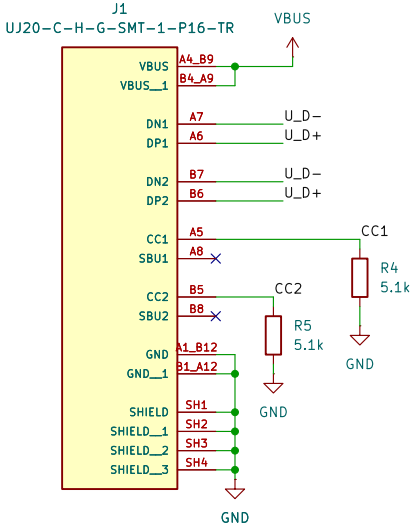


Sheet: /
File: Riddler–Datapack mini.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 1/4

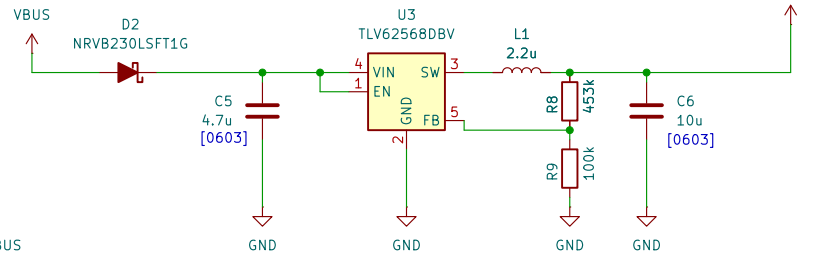
POWER CONNECTORS

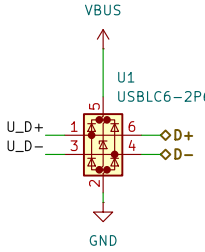


ATTENTION FOR L1:
I HIGHLY RECOMMAND
USING A SHIELDED COIL
TO REDUCE EMI

+3V3 DCDC Buck

DCDC: 3.3V / 1A





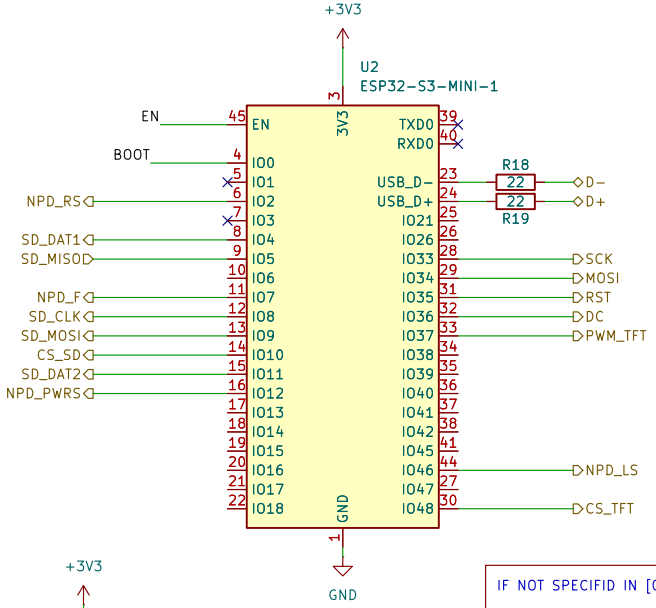
IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Power/
File: power.kicad_sch

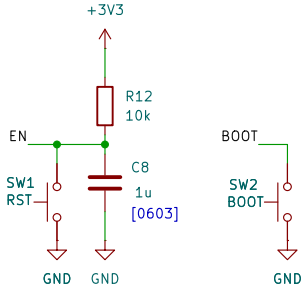
Title:

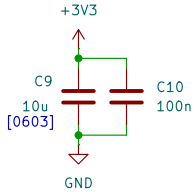
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 2/4

ESP32-S3 N16R8



EN & BOOT





ESP32-S3 chip strapping pins:

GPIO 0.
GPIO 3.
GPIO 45.
GPIO 46.

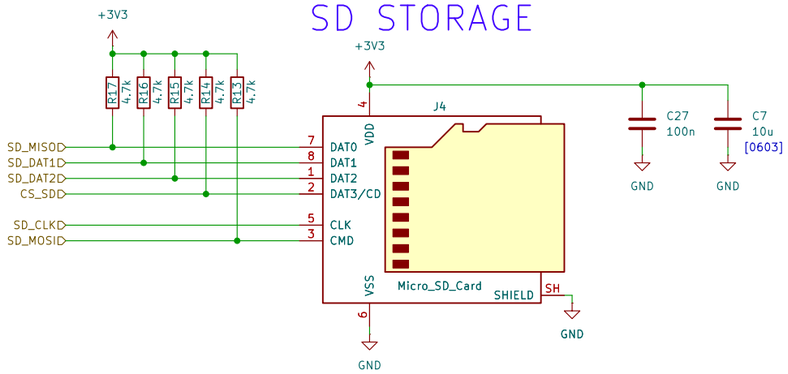
IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Logic/
File: logic.kicad_sch

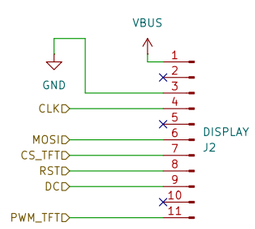
Title:

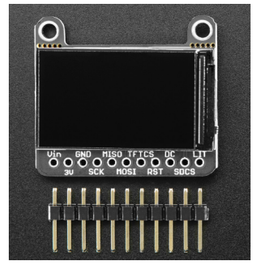
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 3/4

SD STORAGE



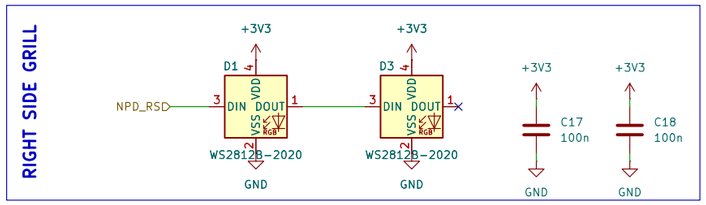
DISPLAY

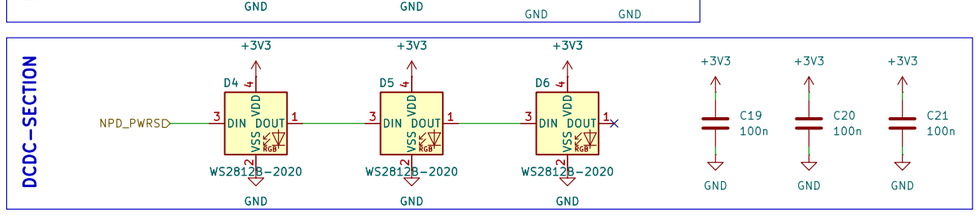


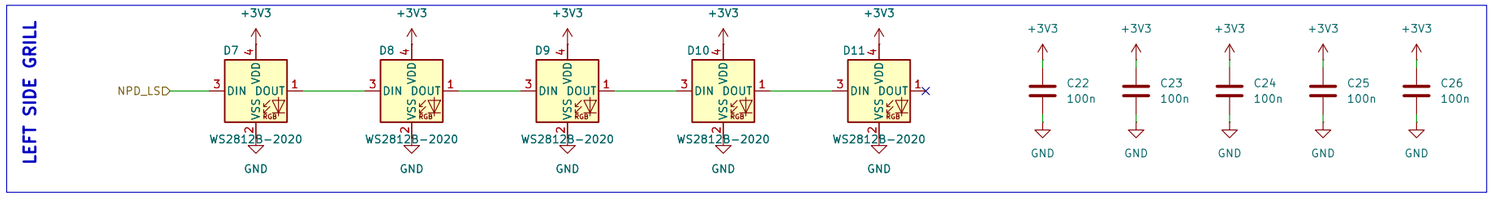


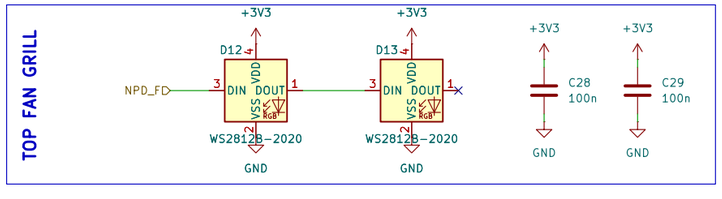
3–5V / Vin – this is the power pin, connect to 3–5VDC – it has reverse polarity protection but try to wire it right!
3V – this is the 3.3V output from the onboard regulator
GND – this is the power and signal ground pin
SCK – this is the SPI clock input pin. Use 3–5V logic level
MISO – this is the SPI Microcontroller In Serial Out pin, it's used for the SD card. It isn't used for the TFT display which is write–only. It is 3.3V logic out (but can be read by 5V logic)
MOSI – this is the SPI Microcontroller Out Serial In pin, it is used to send data from the microcontroller to the SD card and/or TFT. Use 3–5V logic level
TFTCS – this is the TFT SPI chip select pin. Use 3–5V logic level
RST – this is the TFT reset pin. Connect to ground to reset the TFT! It's best to have this pin controlled by the library so the display is reset cleanly, but you can also connect it to the Arduino Reset pin, which works for most cases.
There is an automatic–reset chip connected so it will reset on power–up. Use 3–5V logic level
DC – this is the TFT SPI data or command selector pin. Use 3–5V logic level
SD Card CS / SDCS – this is the SD card chip select, used if you want to read from the SD card. Use 3–5V logic level
LIT – this is the PWM input for the backlight control. It is by default pulled high (backlight on) you can PWM at any frequency or pull down to turn the backlight off. Use 3–5V logic level

NEOPIXELS









IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Peripherals/
File: peripherals.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.5 | Id: 4/4